

1. Given dataset: $X=\{2,4,10,12,3,20,30,11,25\}$

Apply K-Means with $K=2$, initial centroids = 2 and 10. Perform iterations and show final clusters.

- Explain convergence of K-Means
- Prove that K-Means minimizes **within-cluster variance**

2. Given dataset: $X=\{(2,3),(3,4),(4,5),(5,6)\}$

Compute covariance matrix

- Find eigenvalues and eigenvectors
- Project data onto first principal component

3. Given:

$$K(x, z) = (x^T z)^2$$

Find feature mapping $\phi(x)$ for: $x=(x_1, x_2)$